

from Beginner to Master

DevOps를 위한 **Docker** 가상화

Section 1. Docker Essentials - Container

- Section 0: Introduction to Cloud Native Technologies
- **Section 1: Docker Essentials - Container**
- Section 2: Docker Essentials - Image
- Section 3: Docker Network and Storage
- Section 4: Building and Managing Containerized Application
- Section 5: Container Orchestration
- Section 6: Continuous Integration and Continuous Deployment
- Section 7: Docker Security
- Section 8: Logging and Monitoring
- Section 9: Advanced Docker Usage
- Section 10: Capstone Project

Section 1.

Docker Essentials - Container

- Docker와 컨테이너 가상화 기술
- Docker Image
- Docker Container
- Docker Container로 구축하는 IT 인프라

Docker와 컨테이너 가상화 기술

Docker Architecture

- CNCF <https://landscape.cncf.io/>



Docker Architecture

- DevOps Tools

<https://survey.stackoverflow.co/2023/#technology>



Docker Architecture

- 2014년 6월 Docker 1.0 발표
- 컨테이너 기반의 오픈소스 가상화 플랫폼
 - | 일반PC, AWS, Azure, Google cloud 등에서 실행 가능
 - | OS, Backend, Frontend, Database, Message Queue → 컨테이너로 추상화 가능
 - | 2019.11 Mirantis에 인수 (Docker Enterprise)
 - | <https://www.docker.com/blog/updating-product-subscriptions/>

20M+

Monthly Active Users

16M+

Registered Developers

15M+

Docker Hub Repositories

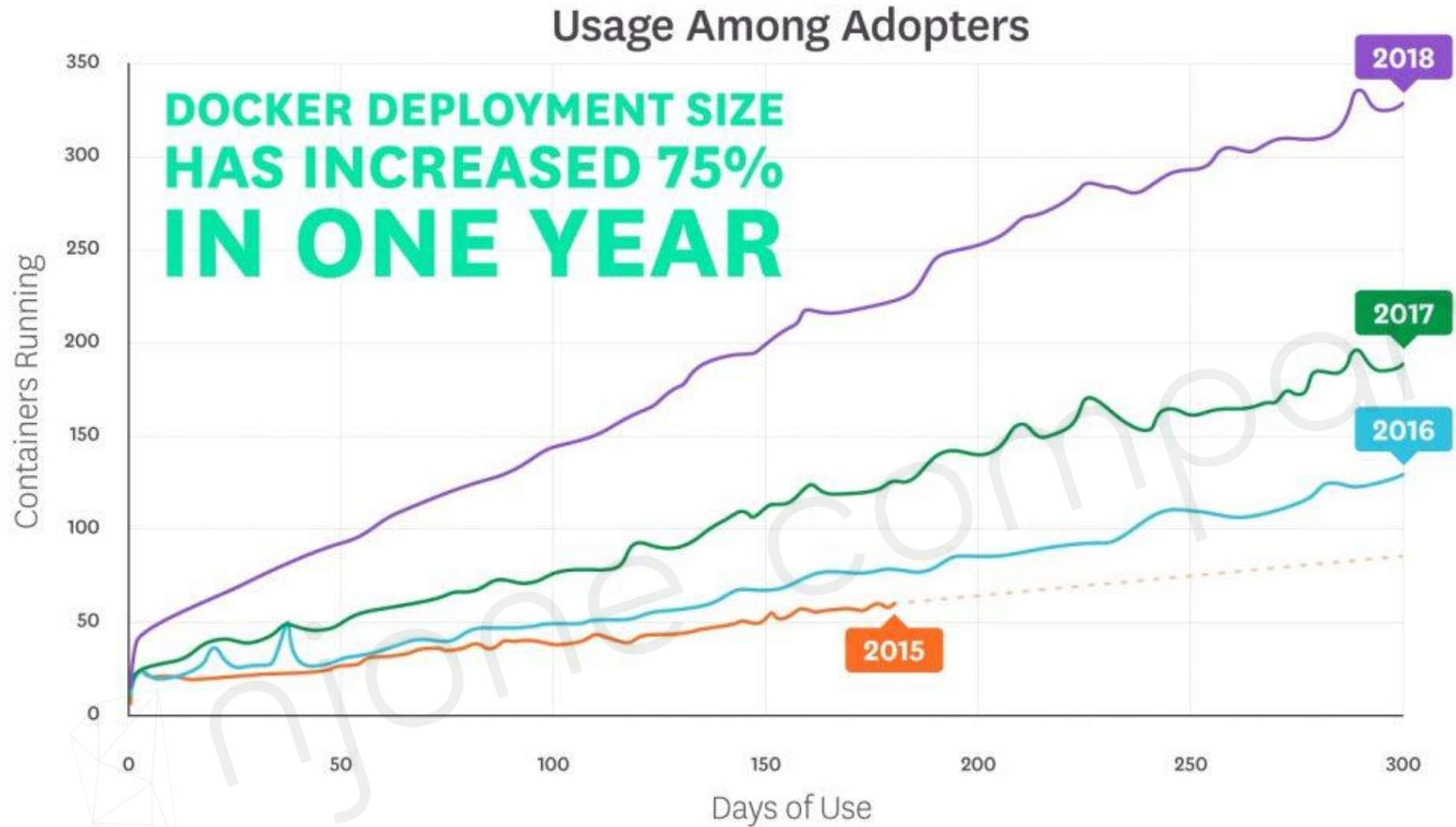
16B+

Monthly Image Pulls

Docker와 컨테이너 가상화 기술

Docker

- 2014
- 컨테이너
- 일반
- OS, B
- 2019.
- <https://>



Docker Architecture

■ 기존 가상화 방식 → OS를 가상화

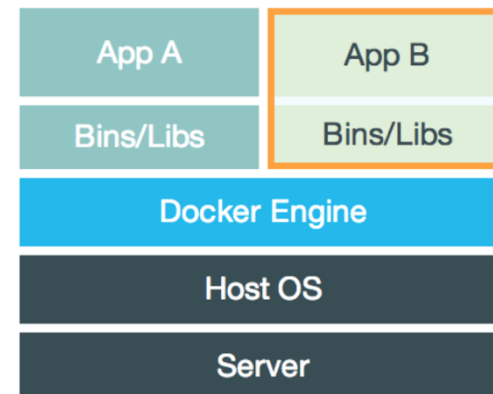
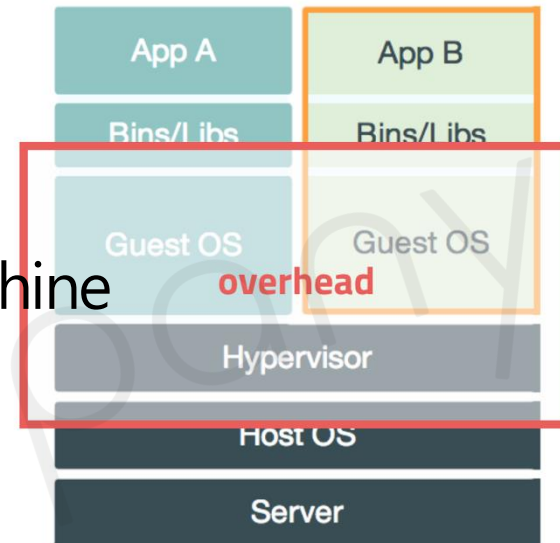
- | VMWare, VirtualBox (Host OS 위에 Guest OS 전체를 가상화)
- | 무겁고 느림

■ CPU의 가상화 기술 이용 방식 → Kernel-based Virtual Machine

- | 전체 OS를 가상화 하지 않음, 호스트 형식에 비해 속도 향상
- | OpenStack, AWS 등의 클라우드 서비스
- | 추가적인 OS는 여전히 필요, 성능 문제

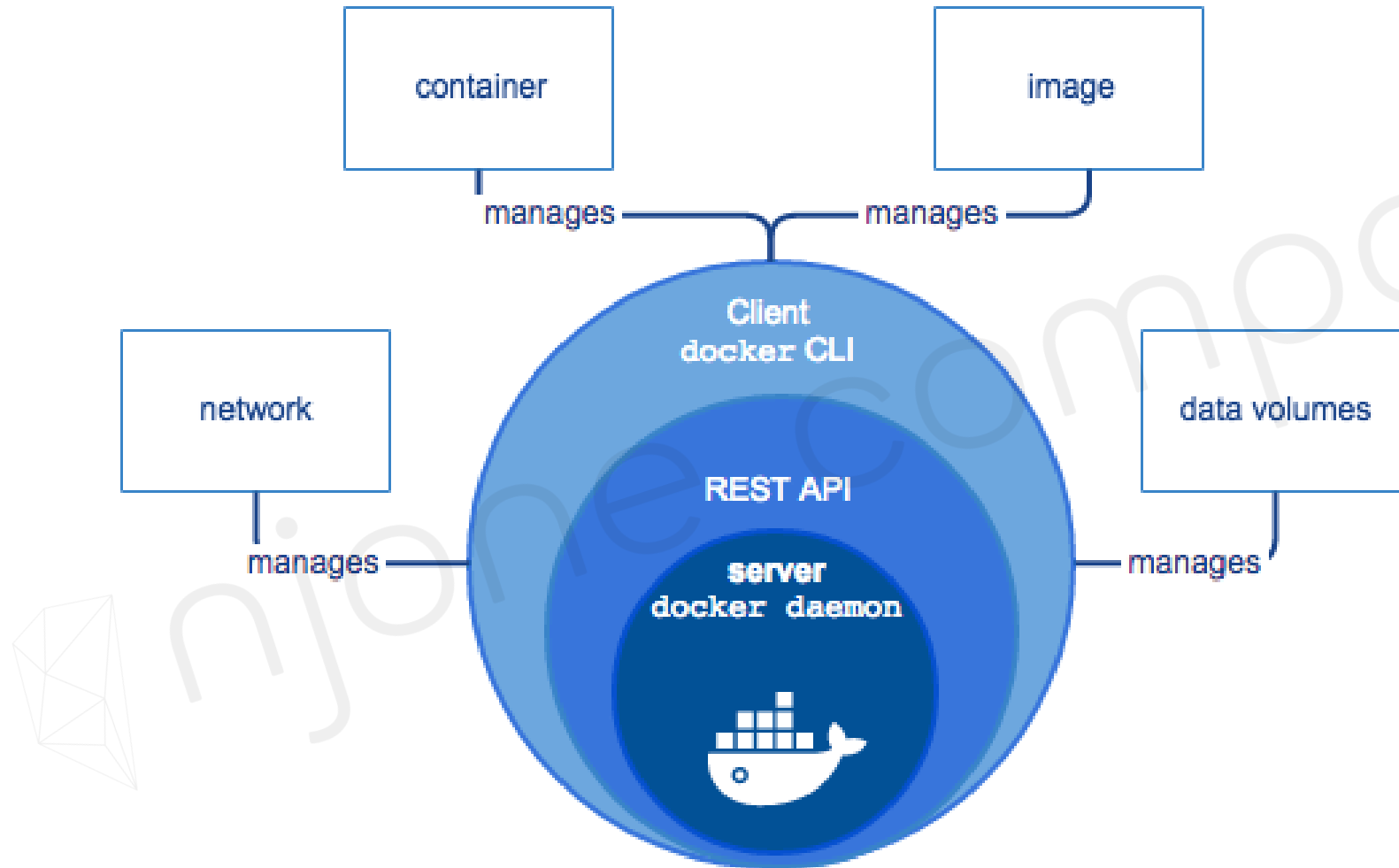
■ 프로세스 격리 → 리눅스 컨테이너

- | CPU나 메모리는 프로세스에 필요한 만큼만 추가로 사용
- | 성능 손실 거의 없음
- | 컨테이너들 사이는 서로 영향을 주지 않음
- | 컨테이너 생성 속도 빠름 (1-2초 내)



Docker Architecture

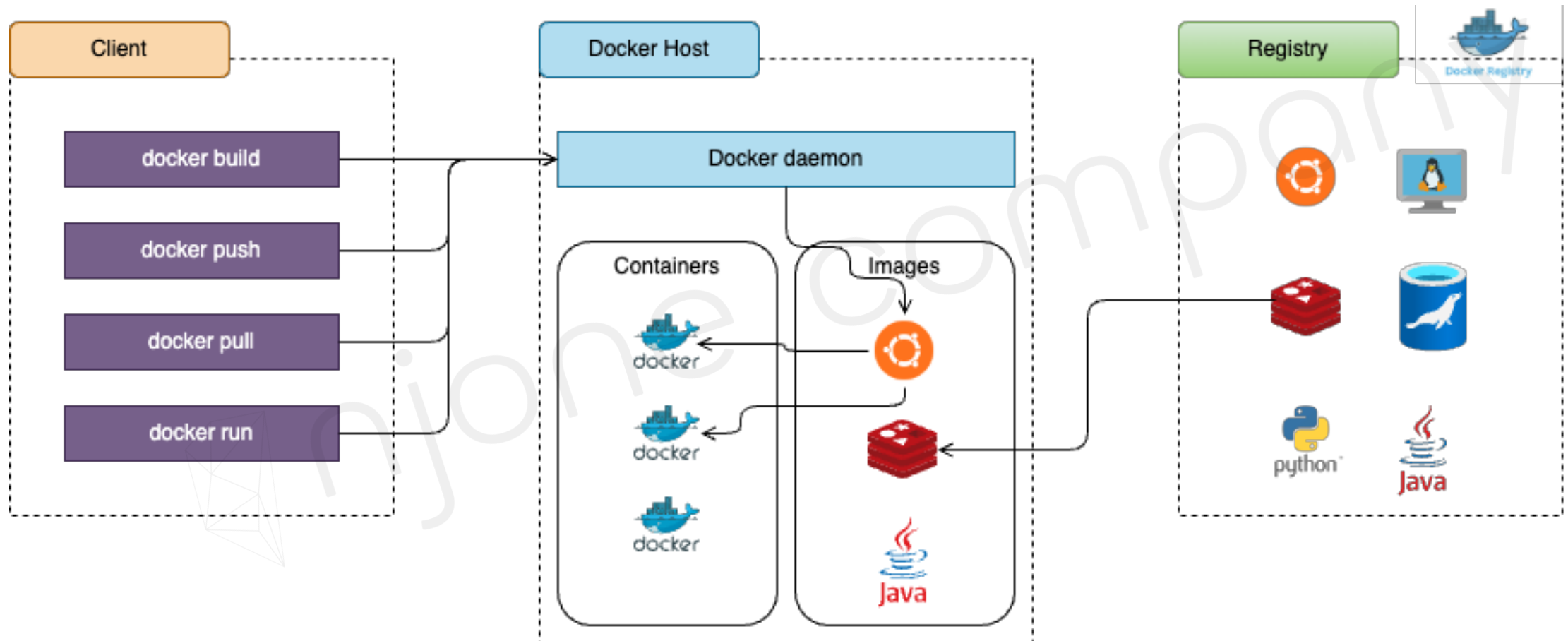
- Docker Engine



Docker vs Container

- 컨테이너 실행에 필요한 파일과 설정 값 등을 포함 ➔ 상태값 X, Immutable

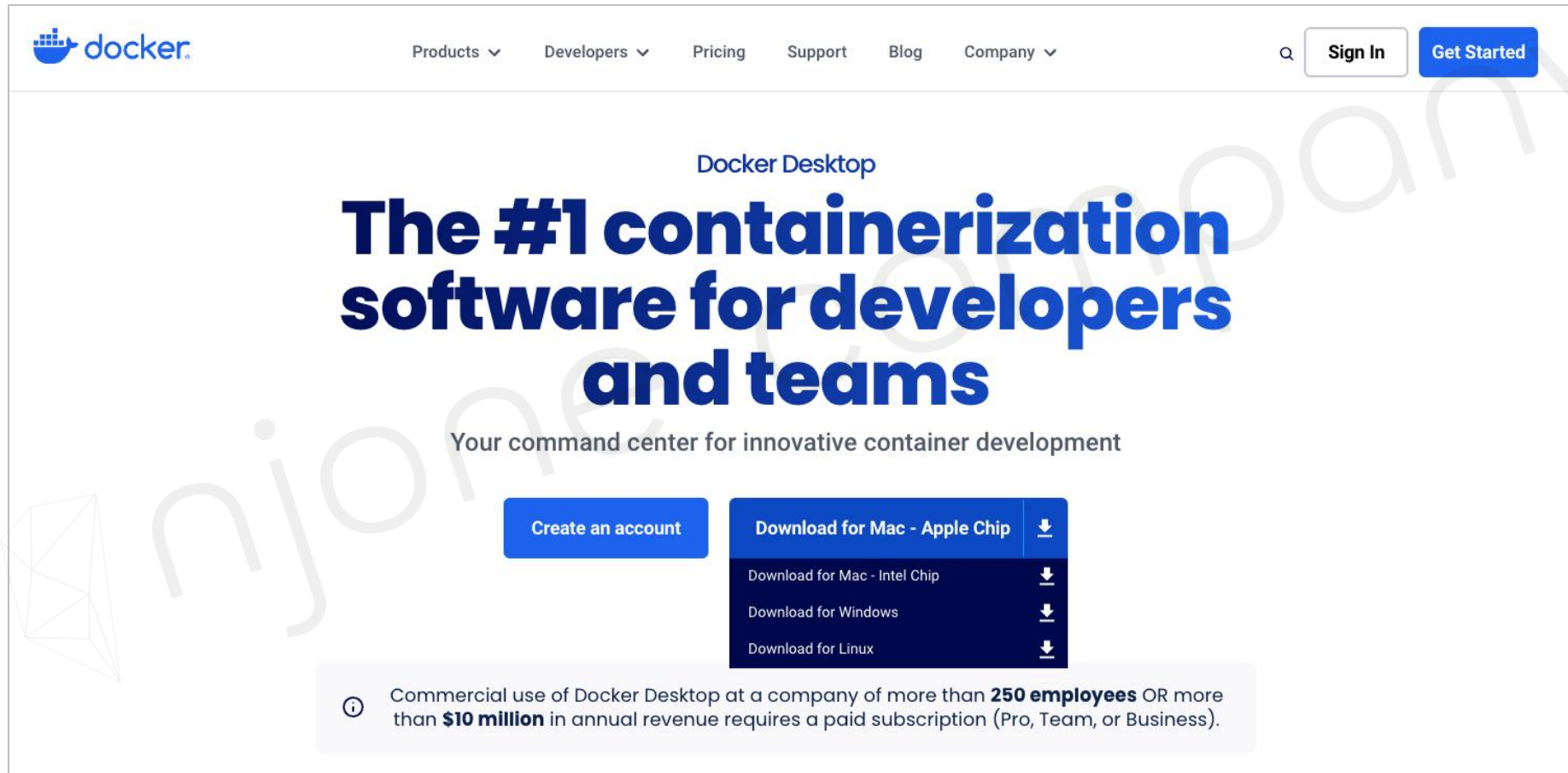
| 실체화 ➔ Container



Docker 실습 환경 구성

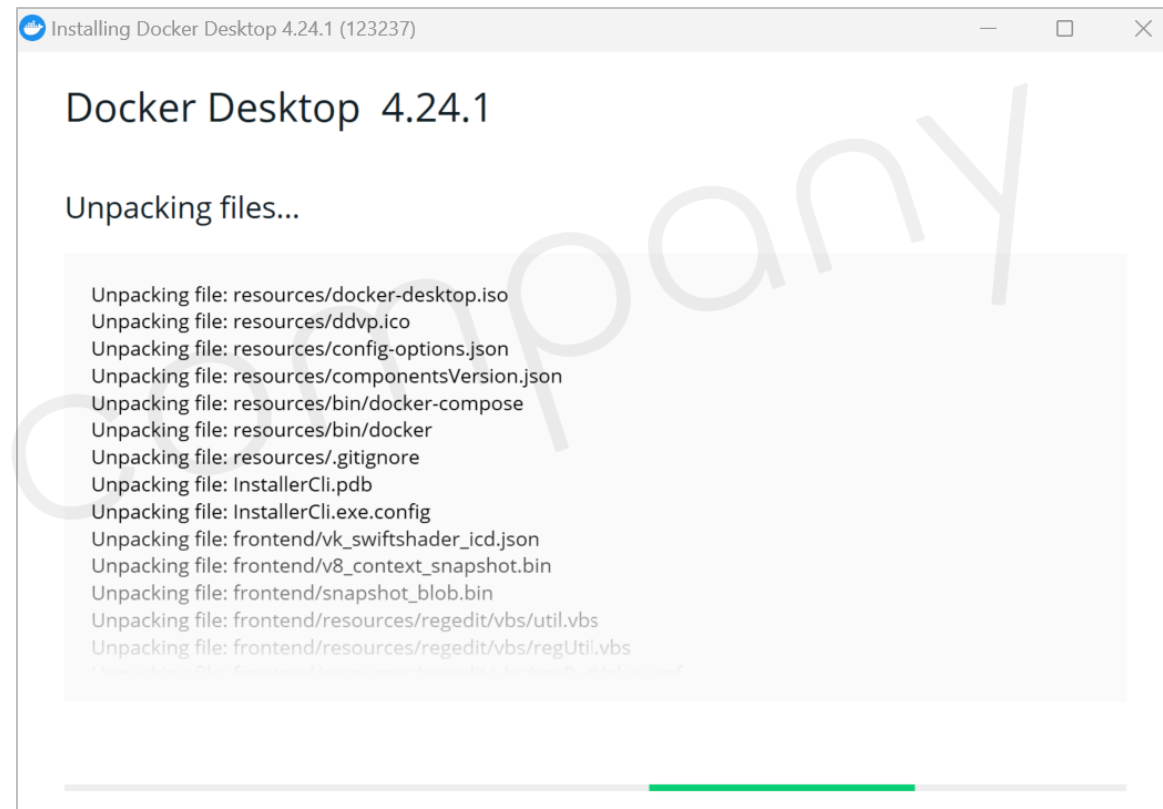
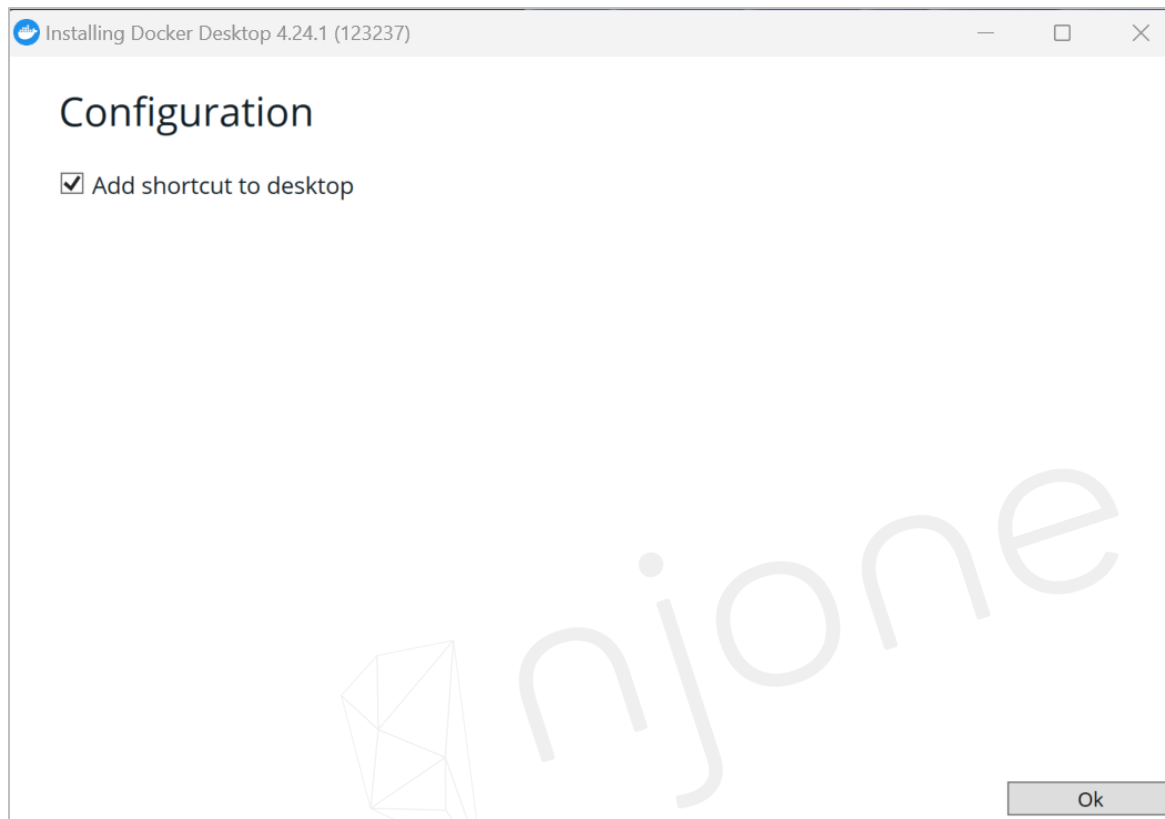
- Docker for Mac/Docker for Windows → Docker Desktop

| <https://www.docker.com/products/docker-desktop>



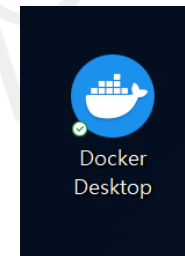
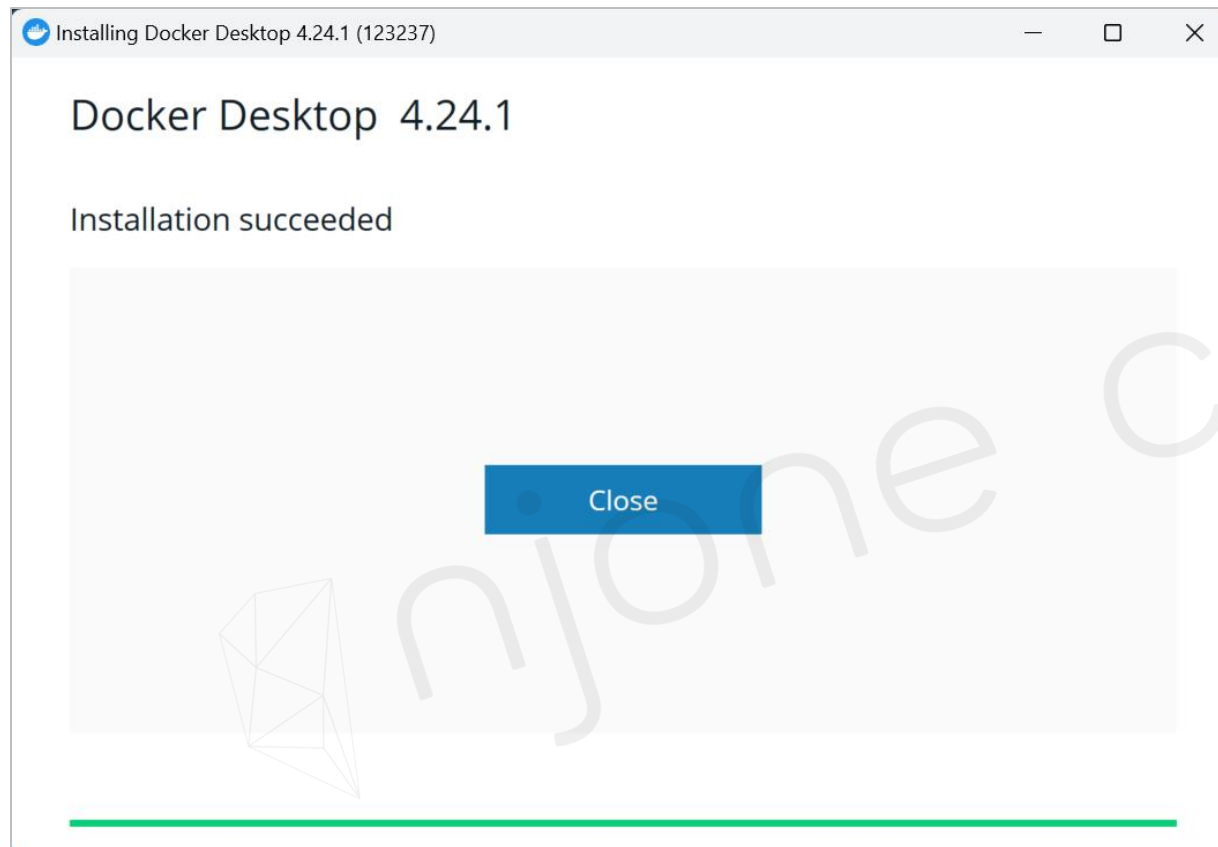
Docker 실습 환경 구성

- Docker for Windows → Docker Desktop



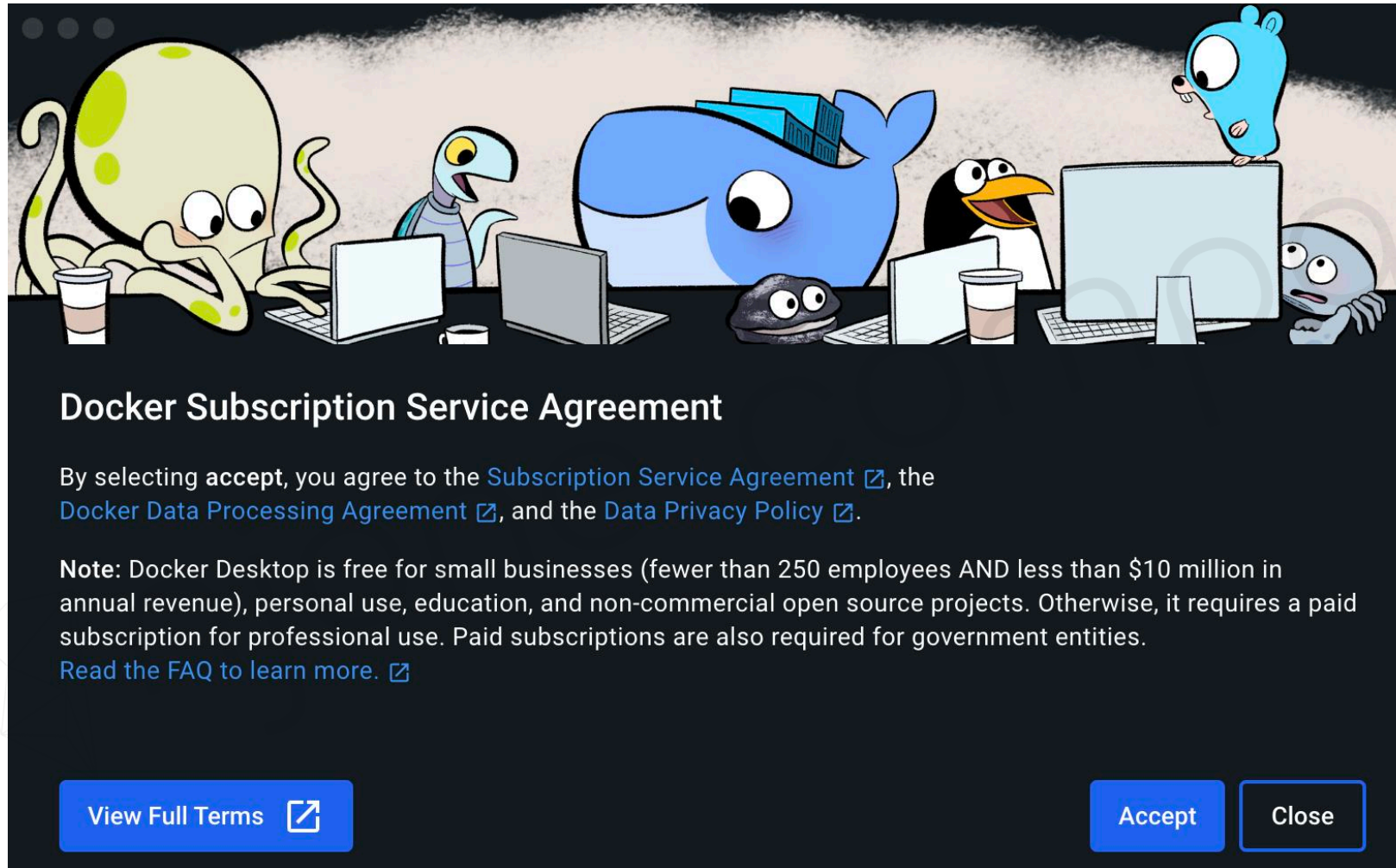
Docker 실습 환경 구성

- Docker for Windows → Docker Desktop



Docker 실습 환경 구성

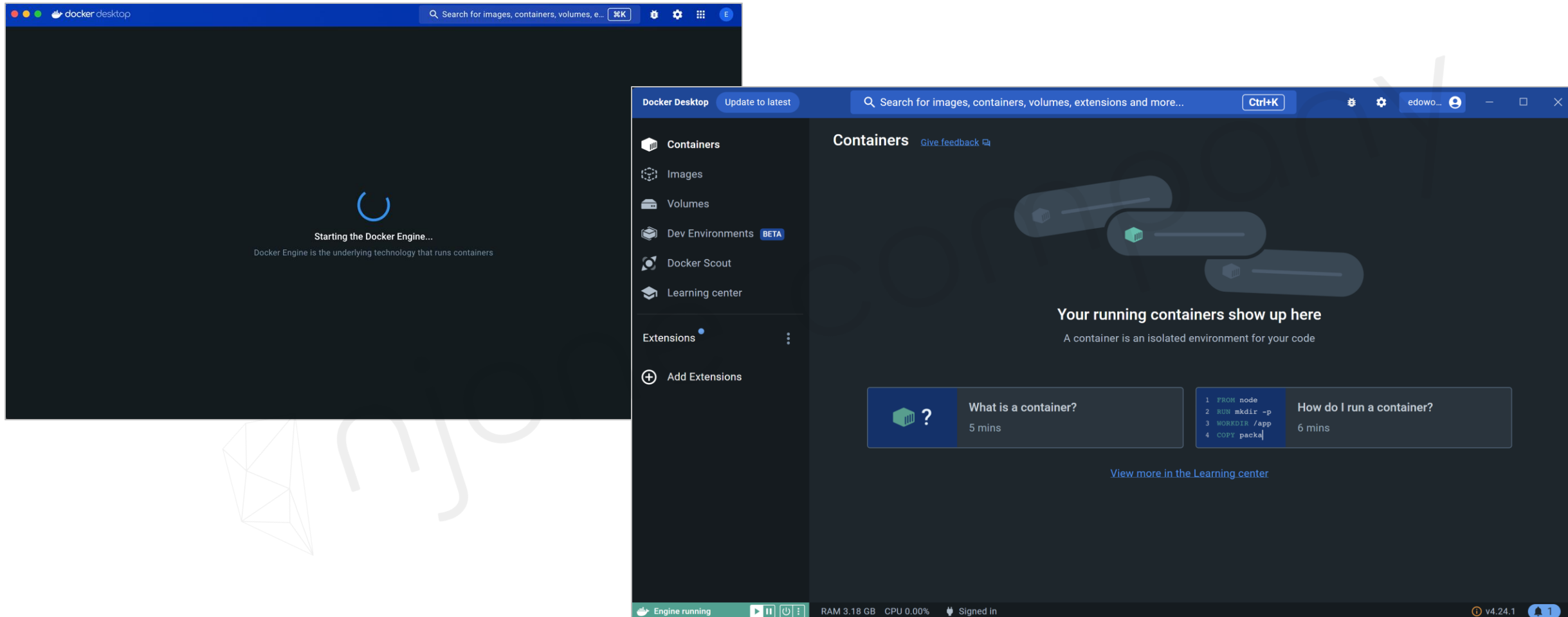
- Docker for Mac/Docker for Windows → Docker Desktop



Docker와 컨테이너 가상화 기술

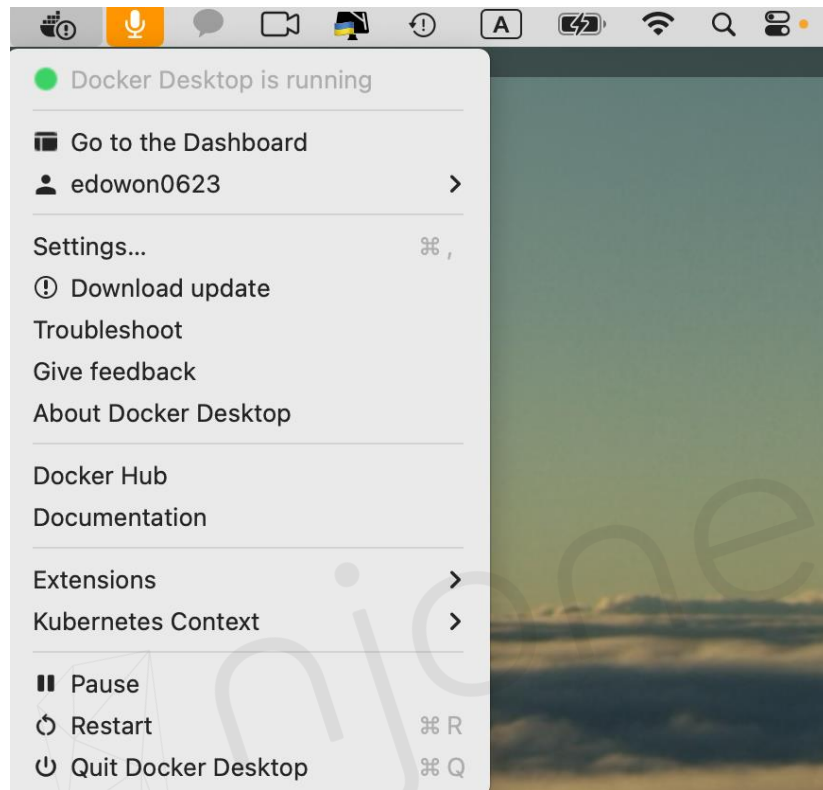
Docker 실습 환경 구성

- Docker for Mac/Docker for Windows → Docker Desktop

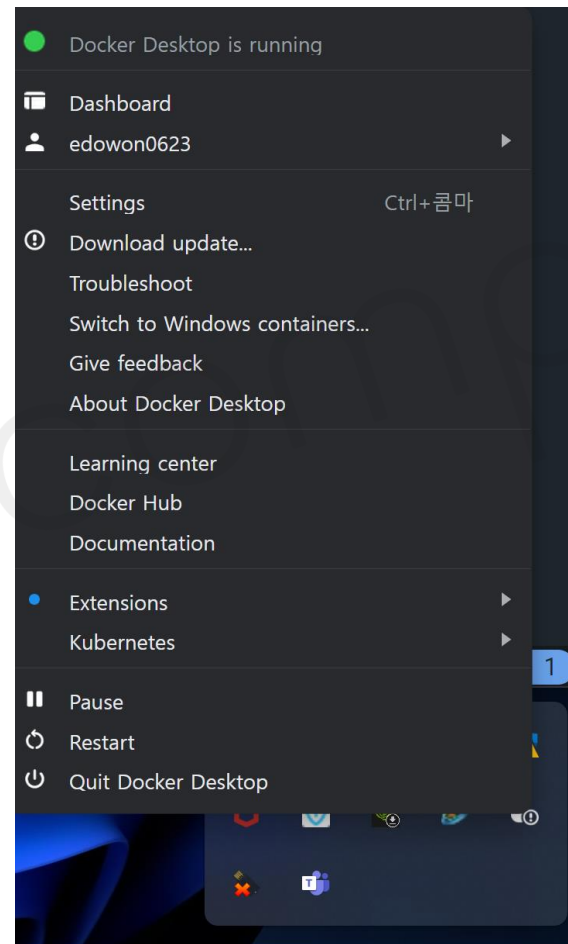


Docker 실행 환경 구성

- Docker for Mac/Docker for Windows → Docker Desktop



Docker for Mac



Docker for Windows

Docker 실습 환경 구성

■ 설치 확인

| \$ docker version

```
bcadmin@hlf03:~$ docker version
```

Client:

Cloud integration: v1.0.35+desktop.13

Version: 26.0.0

API version: 1.45

Go version: go1.21.8

Git commit: 2ae903e

Built: Wed Mar 20 15:14:46 2024

OS/Arch: darwin/arm64

Context: desktop-linux

Server: Docker Desktop 4.29.0 (145265)

Engine:

Version: 26.0.0

API version: 1.45 (minimum version 1.24)

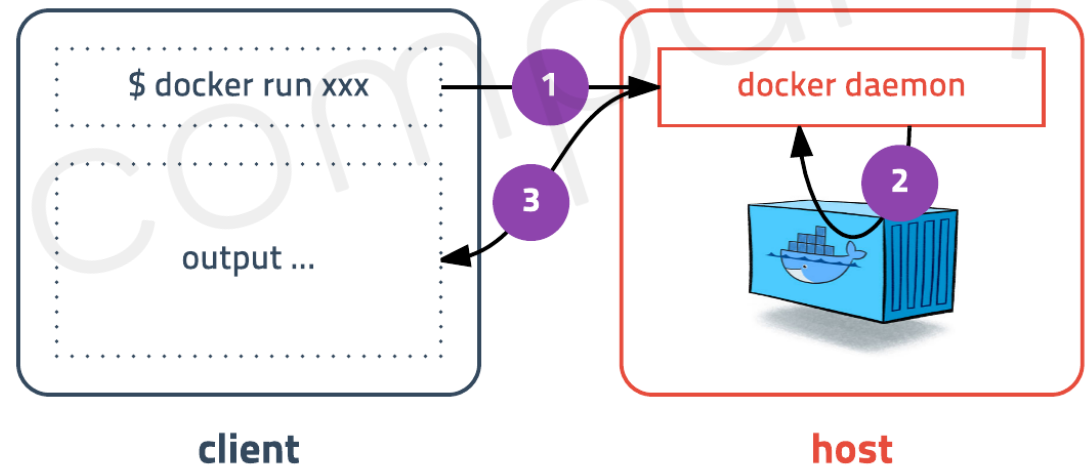
Go version: go1.21.8

Git commit: 8b79278

Built: Wed Mar 20 15:18:02 2024

OS/Arch: linux/arm64

Experimental: false

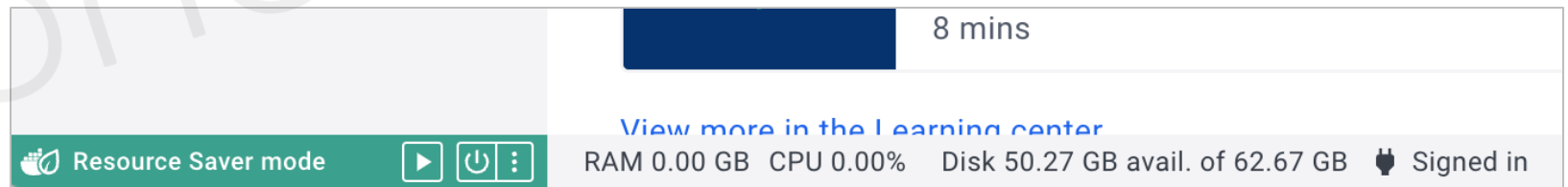
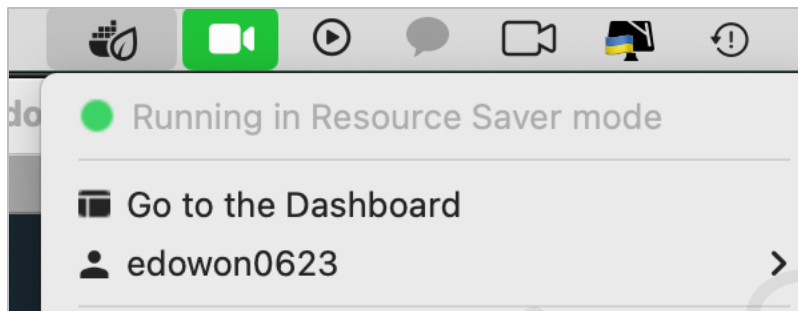


Resource Saver mode

■ Resource Saver

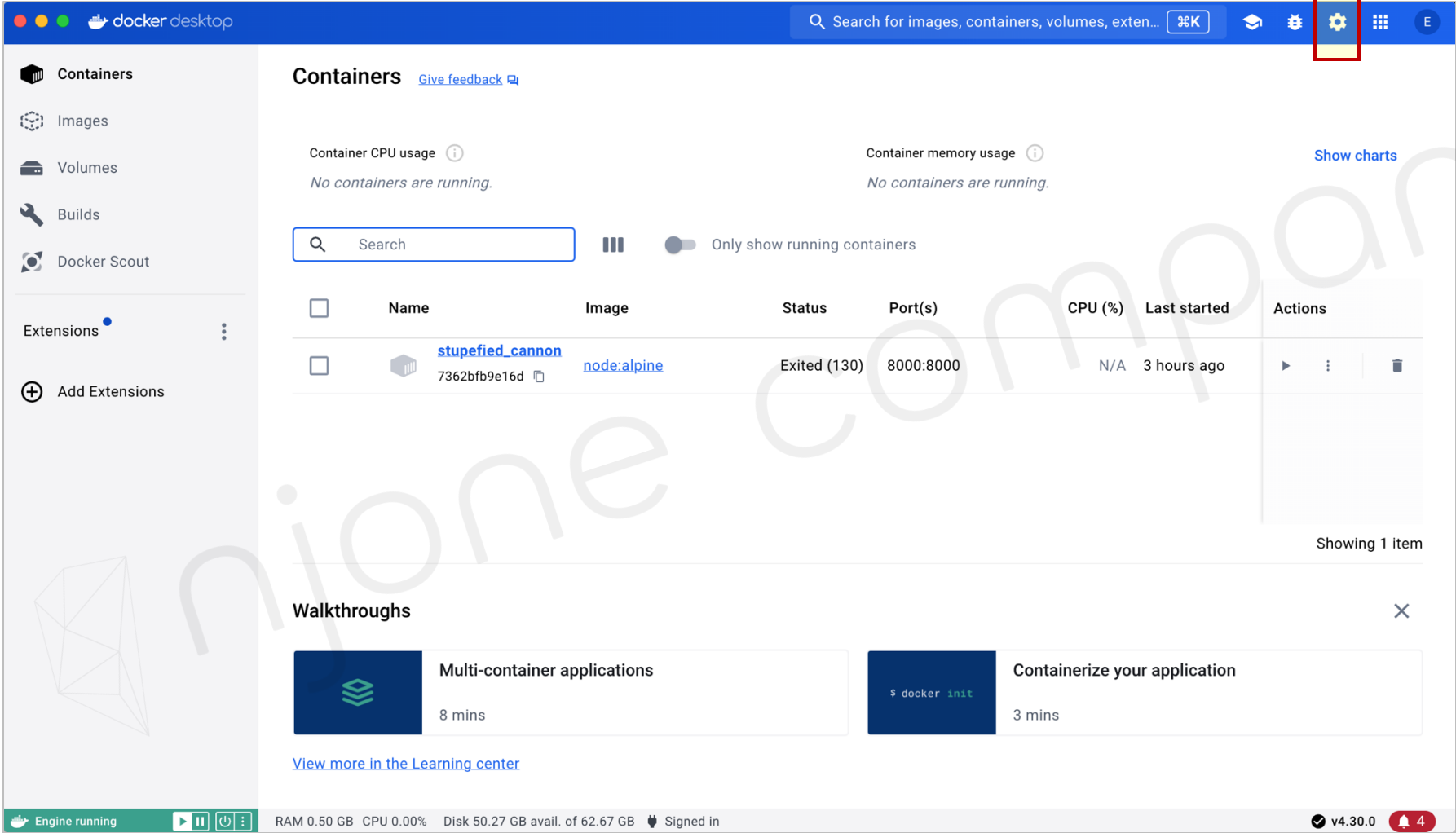
| <https://docs.docker.com/desktop/use-desktop/resource-saver/>

| Docker Desktop version 4.24 부터 적용 → 사용 중인 컨테이너가 없을 때 Docker를 Idle 상태로 운영



Resource Saver mode

■ Resource Saver



The screenshot shows the Docker Desktop application window. In the top right corner of the title bar, the settings gear icon is highlighted with a red rectangle. The main interface displays the 'Containers' tab, which shows that no containers are currently running. A table lists one container, 'stupefied_cannon', which has exited. The bottom status bar indicates that the Docker Engine is running and shows system resource usage.

Containers [Give feedback](#)

Container CPU usage ⓘ
No containers are running.

Container memory usage ⓘ
No containers are running.

[Show charts](#)

Search

Only show running containers

	Name	Image	Status	Port(s)	CPU (%)	Last started	Actions
☐	stupefied_cannon 7362bfb9e16d	node:alpine	Exited (130)	8000:8000	N/A	3 hours ago	▶ ⋮ 🗑️

Showing 1 item

Walkthroughs [×](#)

**Multi-container applications**
8 mins

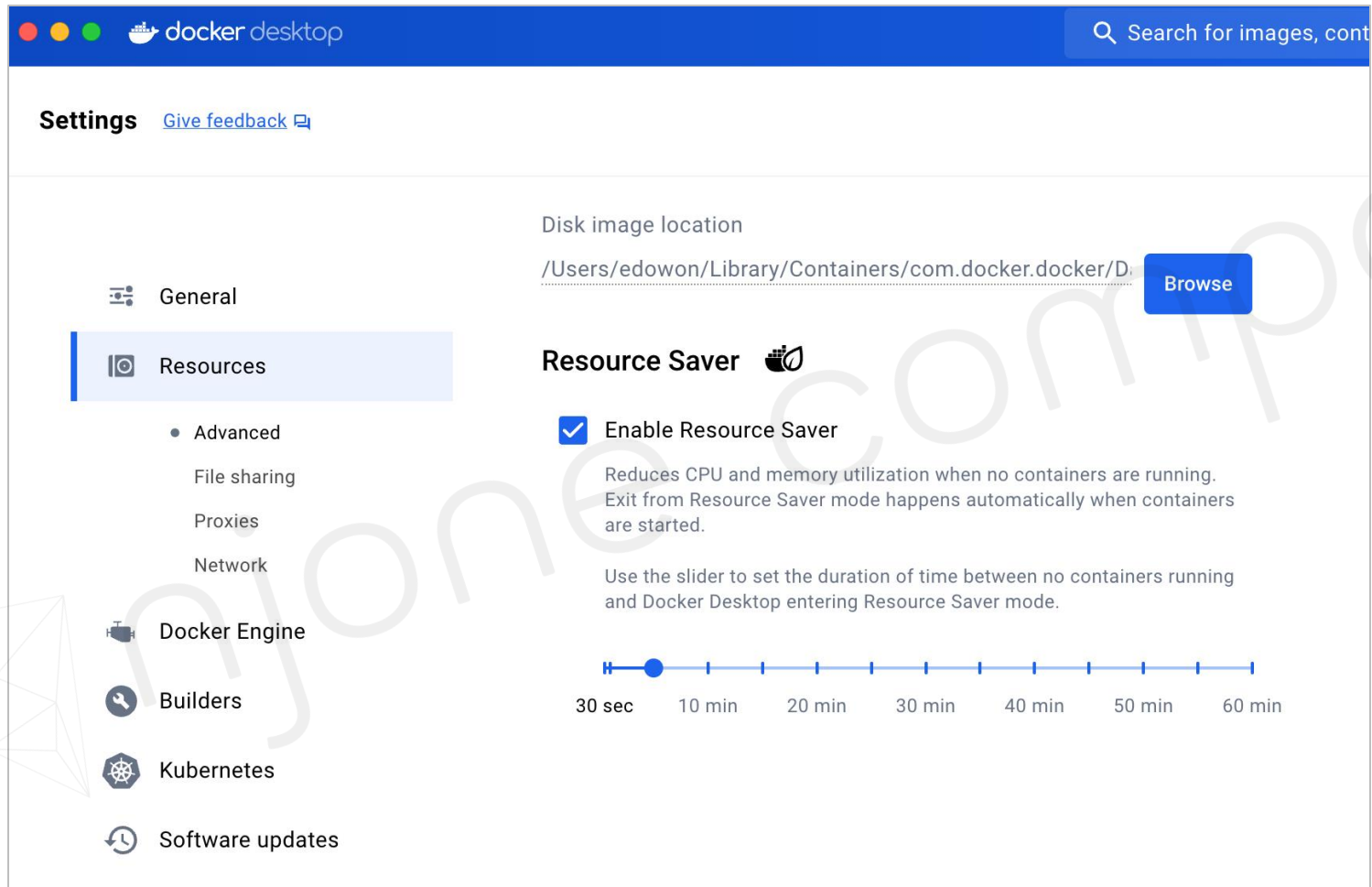
**Containerize your application**
3 mins

[View more in the Learning center](#)

Engine running [▶](#) [⏸](#) [🔌](#) [ⓘ](#) RAM 0.50 GB CPU 0.00% Disk 50.27 GB avail. of 62.67 GB Signed in v4.30.0 4

Resource Saver mode

■ Resource Saver



Docker Image 구조

- Docker Hub에 등록 or Docker Registry 저장소를 직접 만들어 관리
 - | 공개된 도커 이미지는 820만개 이상, 다운로드 수는 100억회 이상
- Layer 저장 방식
 - | 유니온 파일 시스템을 이용 → 여러 개의 Layer를 하나의 파일시스템으로 사용 가능

```
FROM ubuntu:18.04
```

```
COPY . /app
```

```
RUN make /app
```

```
CMD python /app/app.py
```

asd43sdf34ws

asd4sdsf344f

sd345dsf2345

sd4fdsgd57fg

Container Layer

Layer 3 --> CMD python /app/app.py

Layer 2 --> RUN make /app

Layer 1 --> COPY . /app

Base Layer --> FROM ubuntu:18.04

Read & Write Layer

Read Only Layers

Dockerfile

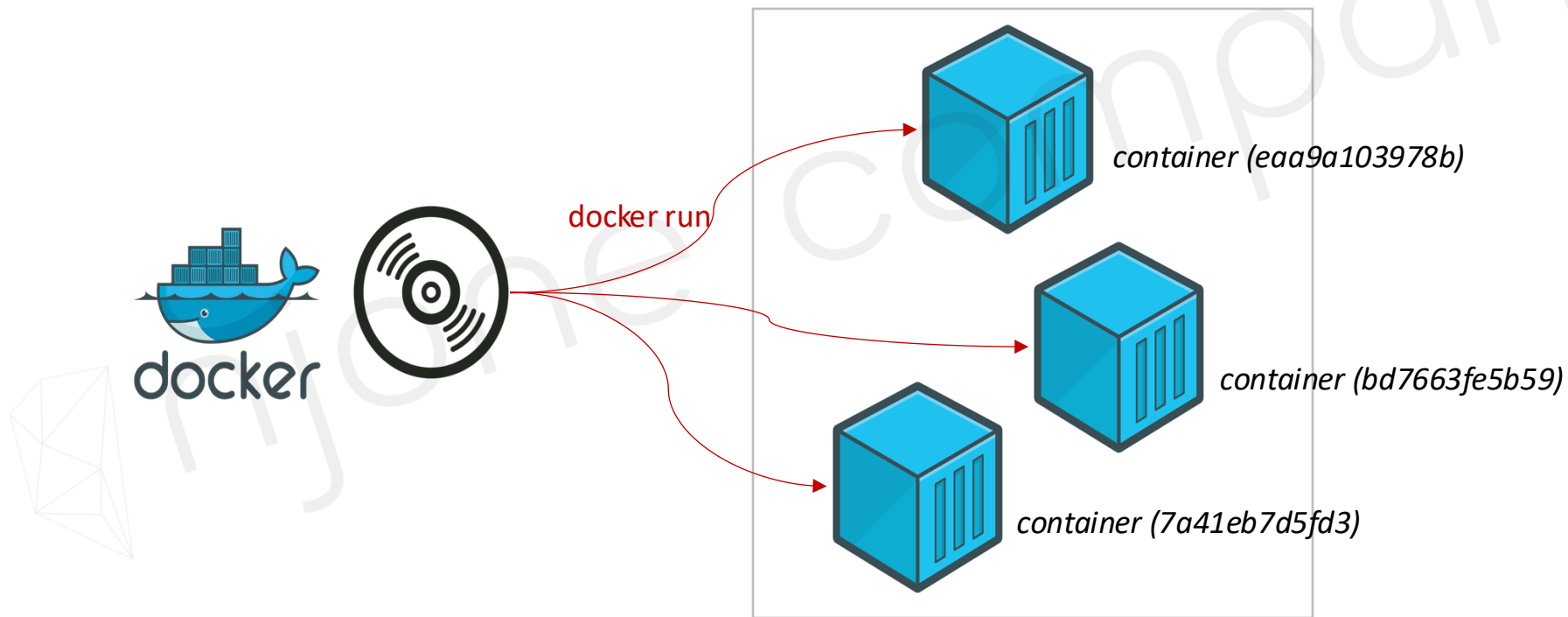
- Docker Image를 생성하기 위한 스크립트 파일
 - | 자체 DSL(Domain-Specific language) 언어 사용 → 이미지 생성과정 기술
 - | 서버에 프로그램을 설치하는 과정을 Dockerfile로 기록, 관리
 - | 소스와 함께 버전 관리가 되며, 누구나 수정 가능

```
FROM node:alpine as builder
WORKDIR /app
COPY ./package.json .
RUN npm install
COPY . .
RUN npm run build

FROM nginx
COPY --from=builder /app/build /usr/share/nginx/html
```

Docker Container 이해

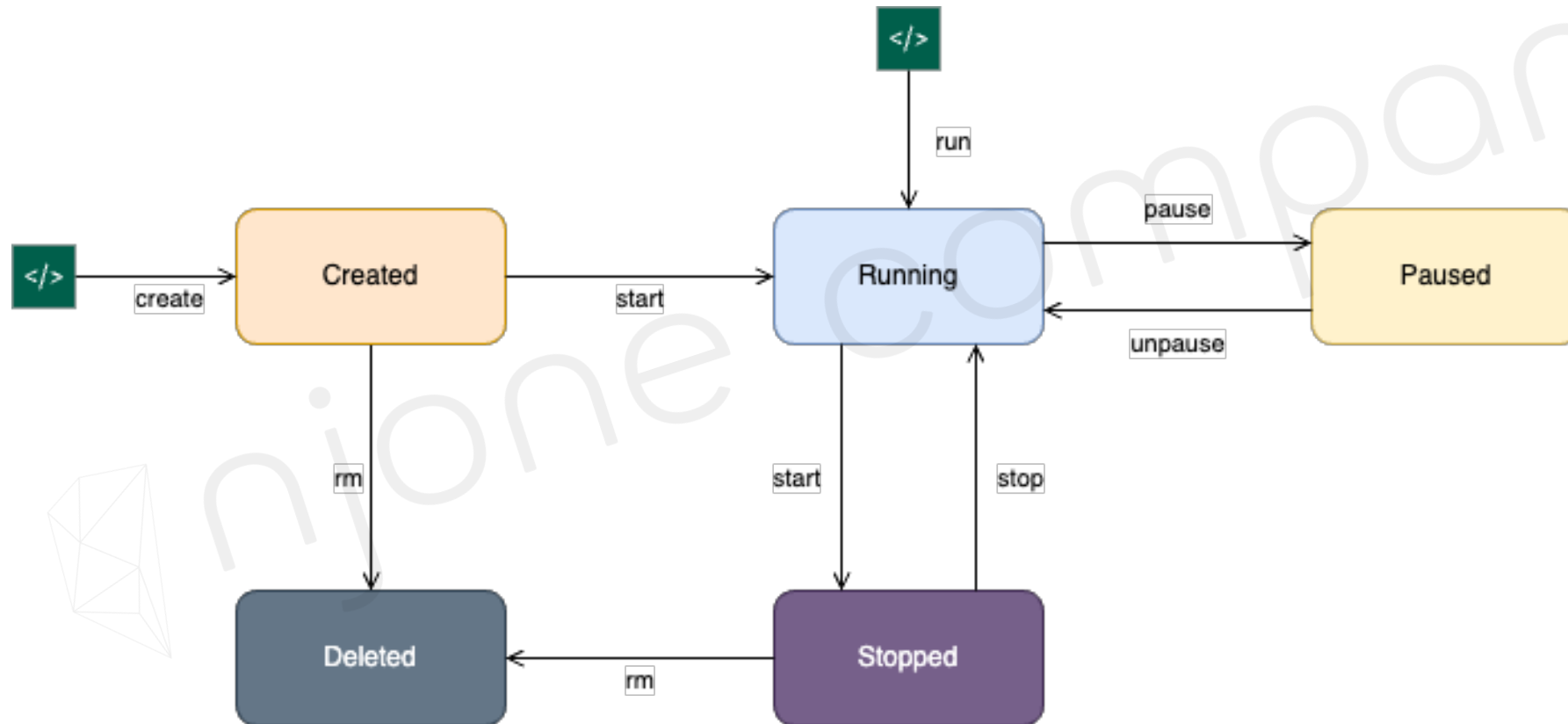
- 도커 이미지를 실행한 상태 → 컨테이너 (=인스턴스)
 - | 격리 된 시스템 자원 및 네트워크를 사용할 수 있는 독립적인 실행 단위
 - | 읽기 전용 상태인 이미지에 변경 된 사항을 저장할 수 있는 컨테이너 계층에 저장



Docker Container 이해

■ Lifecycle

| Docker Container의 실행과 중지에 따른 상태 변화



Docker Container 실행

■ Docker Container 명령어 ①

| \$ docker run **[OPTIONS]** IMAGE[:TAG | @DIGEST] [COMMAND] [ARG...]

옵션	설명
-d, --detach	Detached mode, Background mode
-p, --publish	호스트와 컨테이너의 포트를 연결 (포워딩)
-v, --volume	호스트와 컨테이너의 디렉토리를 연결(마운트)
-e, --env	컨테이너 내에서 사용할 환경변수 설정
--name	컨테이너 이름 설정
--rm	프로세스 종료시 컨테이너 자동 제거
-it	-i와 -t를 동시에 사용한 것으로 터미널 입력을 위한 옵션
-link	컨테이너 연결 [컨테이너명:별칭]

Docker Container 실행

■ Docker Container 명령어 ①

| 실습) ubuntu 16.04 container 생성하고 컨테이너 내부 접속

```
$ docker run ubuntu:16.04
```

```
[admin@centos7 ~]$ docker run ubuntu:16.04
Unable to find image 'ubuntu:16.04' locally
16.04: Pulling from library/ubuntu
35b42117c431: Pull complete
ad9c569a8d98: Pull complete
293b44f45162: Pull complete
0c175077525d: Pull complete
Digest: sha256:a4d8e674ee993e5ec88823391de828a5e
Status: Downloaded newer image for ubuntu:16.04
```

Docker Container 실행

■ Docker Container 명령어 ①

| 컨테이너는 프로세스이기 때문에 실행 중인 프로세스가 없으면 컨테이너는 종료 됨

```
$ docker run --rm -it ubuntu:16.04 /bin/bash
```

```
[admin@centos7 ~]$ docker run --rm -it ubuntu:16.04 /bin/bash
root@b0eac8e53eaa:/# cat /etc/issue
Ubuntu 16.04.6 LTS 🐧 🐧

root@b0eac8e53eaa:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc
```

Docker Container 실행

■ Docker Container 명령어 ①

| 실습) MySQL 5.7 버전을 도커로 실행 https://hub.docker.com/_/mysql

```
$ docker run -d -p 3306:3306 -e MYSQL_ALLOW_EMPTY_PASSWORD=true --name mysql mysql:5.7
```

```
bcadmin@hlf03:~$ docker run -d -p 3306:3306 -e MYSQL_ALLOW_EMPTY_PASSWORD=true --name mysql mysql:5.7
Unable to find image 'mysql:5.7' locally
5.7: Pulling from library/mysql
f17d81b4b692: Pull complete
c691115e6ae9: Pull complete
41544cb19235: Pull complete
254d04f5f66d: Pull complete
4fe240edfdc9: Pull complete
0cd4fcc94b67: Pull complete
8df36ec4b34a: Pull complete
b8edeb9ec9e2: Pull complete
2b5adb9b92bf: Pull complete
5358eb71259b: Pull complete
e8d149f0c48f: Pull complete
Digest: sha256:42bab37eda993e417c5e7d751f1008b653c3fd85ad6aa416a519f161
Status: Downloaded newer image for mysql:5.7
842ff7eb6799b714050615ce505c1f96625343c0589f5d3124e2ec50c852b0f7
```

```
$ docker exec -it mysql bash
```

```
bcadmin@hlf03:~$ docker exec -it mysql bash
root@842ff7eb6799:/# mysql -h127.0.0.1 -uroot
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 2
Server version: 5.7.24 MySQL Community Server (GPL)

Copyright (c) 2000, 2018, Oracle and/or its affiliates. All rights reserved.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.00 sec)

mysql>
```

Docker Container 실행

■ Docker Container 명령어 ②

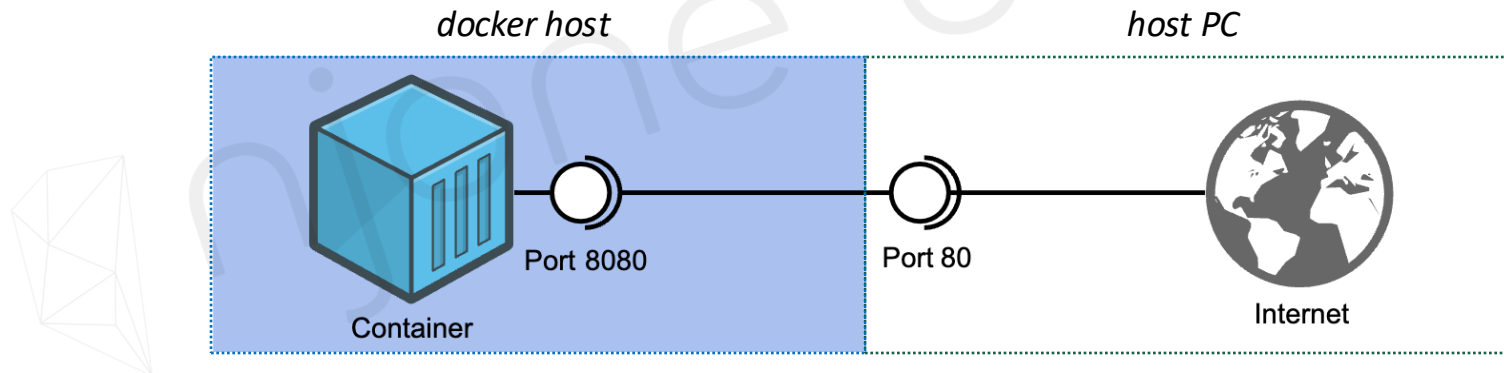
```
$ docker container ls [OPTIONS] ( = docker ps )  
$ docker container stop [OPTIONS] CONTAINER [CONTAINER ...]  
$ docker container rm [OPTIONS] CONTAINER [CONTAINER ...]  
$ docker container logs ${CONTAINER_ID}  
    ex) docker logs -f ${CONTAINER_ID}  
$ docker container exec [OPTIONS] CONTAINER COMMAND [ARG...]  
    ex) docker exec -it mysql /bin/bash  
$ docker container inspect ${CONTAINER_ID}  
$ docker image ls [OPTIONS] [REPOSITORY[:TAG]]  
$ docker image rm [OPTIONS] IMAGE [IMAGE ...]  
$ docker image pull [OPTIONS] NAME[:TAG|@DIGEST]  
    ex) docker pull ubuntu:16.04  
$ docker system prune
```

Docker Container 실행

■ Port Mapping

- | 도커 컨테이너에서 사용하고자 하는 Port를 자유롭게 설정 가능
- | 호스트 시스템에서 도커 컨테이너 Port를 사용하기 위해서는 Port Mapping 필요
- | \$ docker run -p **host_port:container_port** <IMAGE NAME>

```
$ docker run -p 80:8080 <IMAGE NAME>
```



Docker Container 실행

■ Port Mapping

| 실습) 다음과 같은 명령어로 nginx 서버를 기동하였을 때, host PC에서의 접속 방법은?

```
$ docker run -p 80:80 nginx
```



Docker Container 실행

■ Detached Mode

| 컨테이너를 백그라운드 모드를 실행

```
$ docker run -d -p 80:80 nginx
```

```
▶ docker run -d -p 80:80 nginx
2fc5d862942c9898b63dd914161fbdab90a1f7018dff0c025f113a47f5e5724a
(base) edowon ~
```

■ Container List

| 컨테이너를 백그라운드 모드를 실행

```
$ docker ps
```

```
▶ docker ps
CONTAINER ID   IMAGE
2fc5d862942c   nginx
```


Docker Container 실행

■ Remove Container

컨테이너 종료와 함께 컨테이너 삭제

```
$ docker run --rm -d -p 80:80 nginx
```

```
▶ docker stop 64fe72
```

```
64fe72
```

```
(base) edowon ~
```

```
▶ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
--------------	-------	---------	---------	--------	-------

```
▶ docker run --rm -d -p 80:80 nginx
```

```
64fe724dff0efa5f05e1195132baee36cbf625ece1c20c2a78b7feb2609c6e3f
```

```
(base) edowon ~
```

```
▶ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
64fe724dff0e	nginx	"/docker-entrypoint...."	10 seconds ago	Up 9 seconds	0.0.0.0:80->80/tcp

Docker Container 실행

- Docker Container에 명령어 전달 → 실행

| 실행 된 도커 컨테이너 내부에 명령어 실행

```
$ docker exec <CONTAINER_ID> <COMMAND>
```

```
$ docker exec 6sdkjh ls
```

```
$ docker exec -it 6sdkjh bash
```

Docker로 데이터베이스 실행

- 실습) MySQL 5.7 버전을 도커로 실행 https://hub.docker.com/_/mysql

```
$ docker run -d -p 3306:3306 -e MYSQL_ALLOW_EMPTY_PASSWORD=true --name mysql mysql:5.7
```

- 실습) MySQL Client로 접속 테스트

| MySQL Workbench <https://dev.mysql.com/downloads/workbench/>

| HeidiSQL <https://www.heidisql.com/>

| DataGrip <https://www.jetbrains.com/datagrip/>

Docker Container로 구축하는 IT 인프라

Docker로 Node.js 실행

- 실습) 아래 Github의 Repository를 Clone

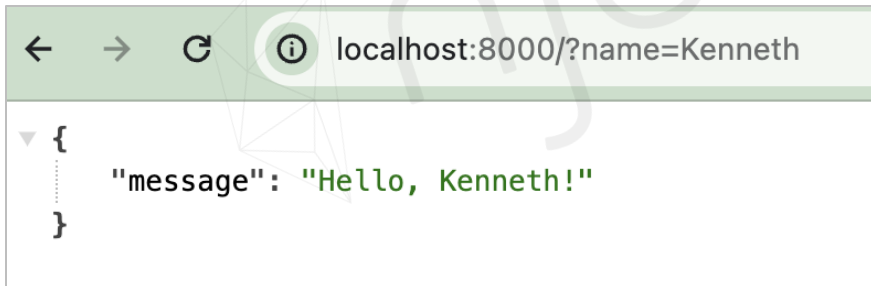
| <https://github.com/joneconsulting/devops-docker>

```
$ cd devops-docker/my-nodejs
```

```
$ docker run -it -v ./docker_nodejs:/app -p 8000:8000 node:alpine sh
```

```
/ # cd app
```

```
/ # node app.js
```



8000:8000



node:alpine